

A Data-Driven Behavioral Analysis to Accelerate Annual Membership Conversions



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Executive Summary

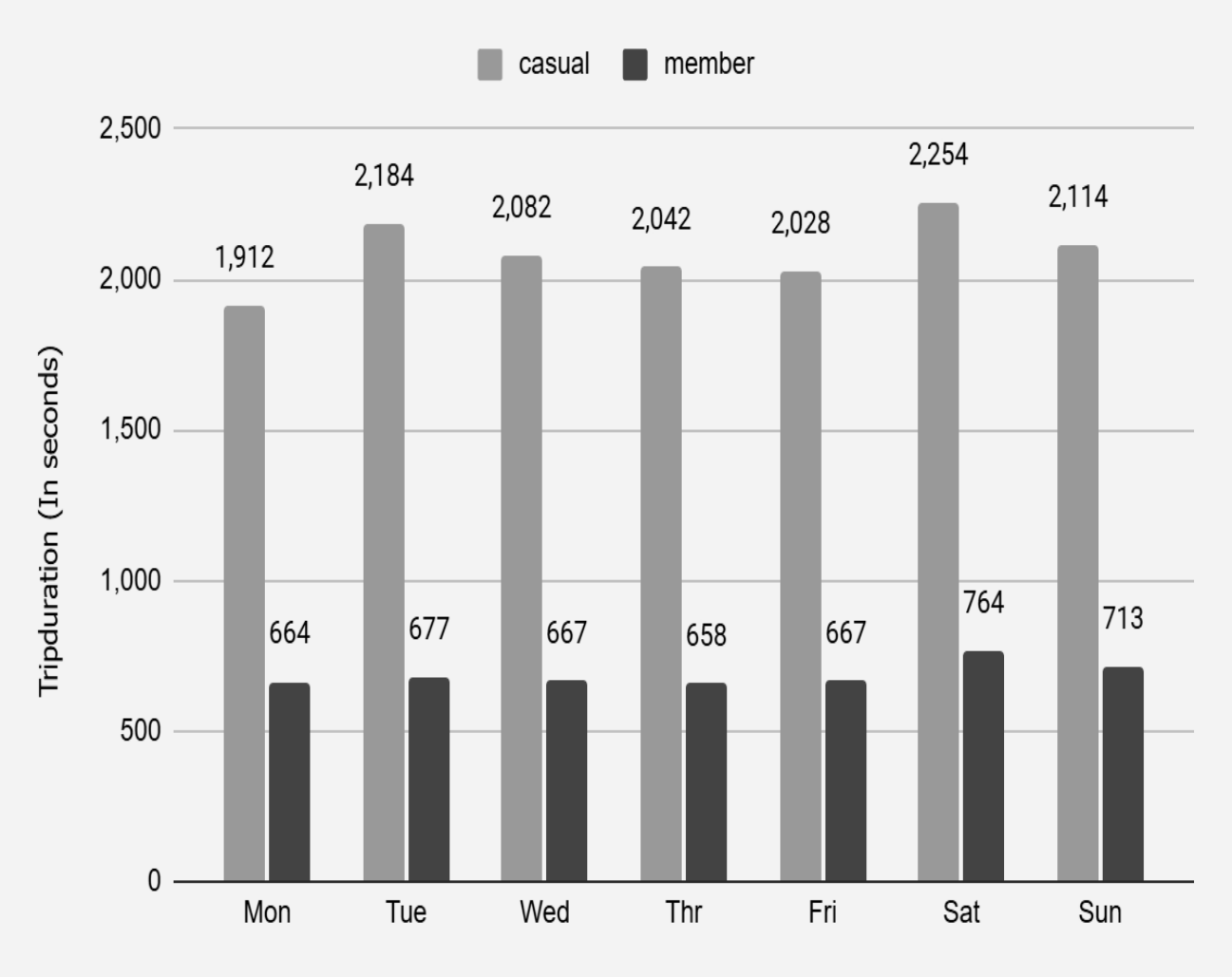
- *To sustain growth, Cyclistic must convert existing casual riders into long-term annual members.*
- *Identify **casual riders** and **annual members** use Cyclistic bikes differently?*

Data Methodology & Integrity

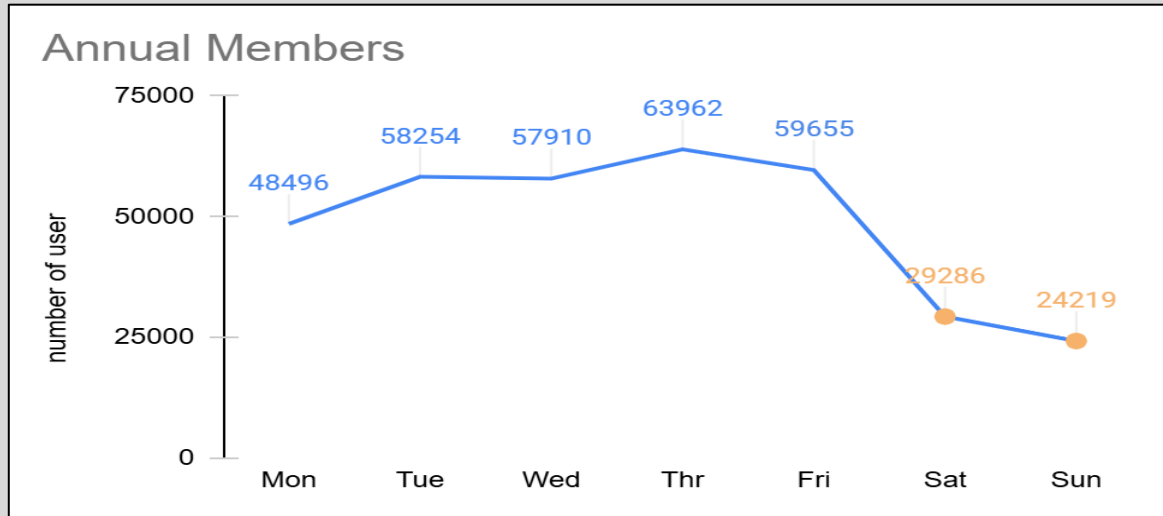
- Source : Cyclistic historical trip data
Divvy_2019 and 2020 Q1
(<https://divvy-tripdata.s3.amazonaws.com/index.html>)
- **Data Cleaning Highlight:**
 - Removed system anomalies, maintenance test runs, and accidental trips (trips that have same start and end station and trip duration 1 ~ 10 seconds)
 - All trip duration are based on a daily basis, removed trip that has > 24 hrs.
 - Add age_group column on 2019 dataset.

Members: Average trip duration is highly efficient at 687 sec ~**11.1 minutes**.

Casuals: Average trip duration is a leisurely 2,088 ~**34.6 minutes**.



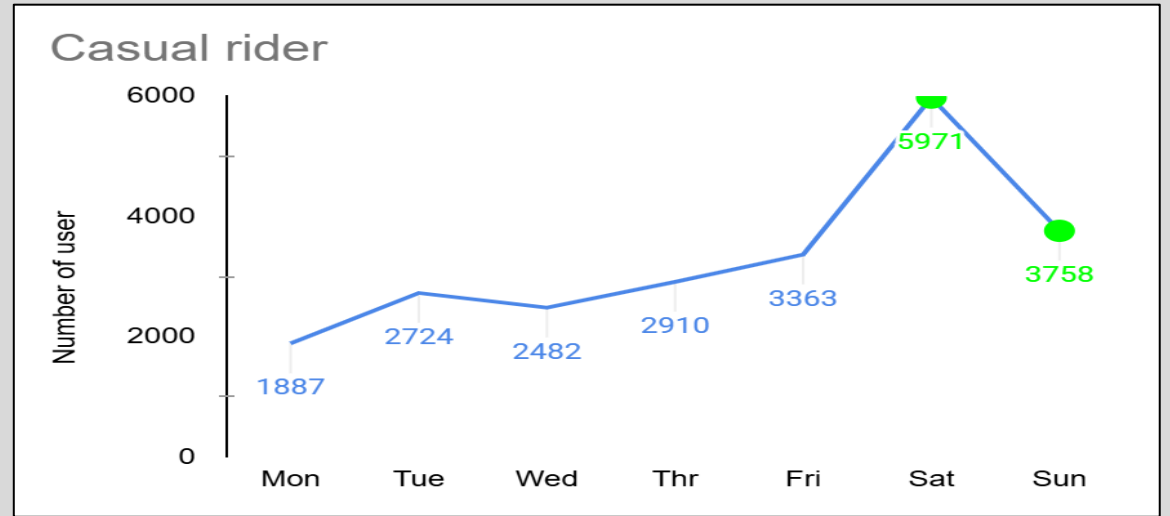
The Commuter



- Typical office day (Stable on Mon – Fri)
- Point -to-point Work commute



The Explorer

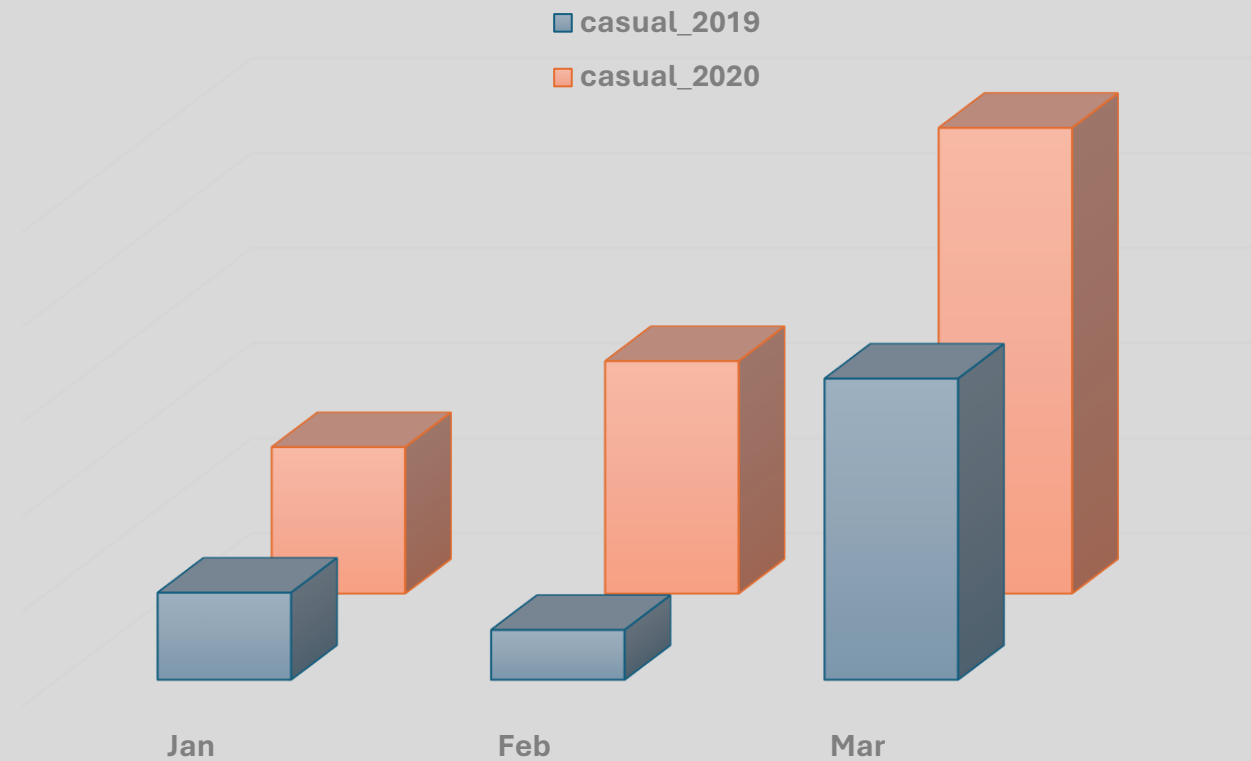


- Extended weekend leisures
- Exploring and taking scenic routes



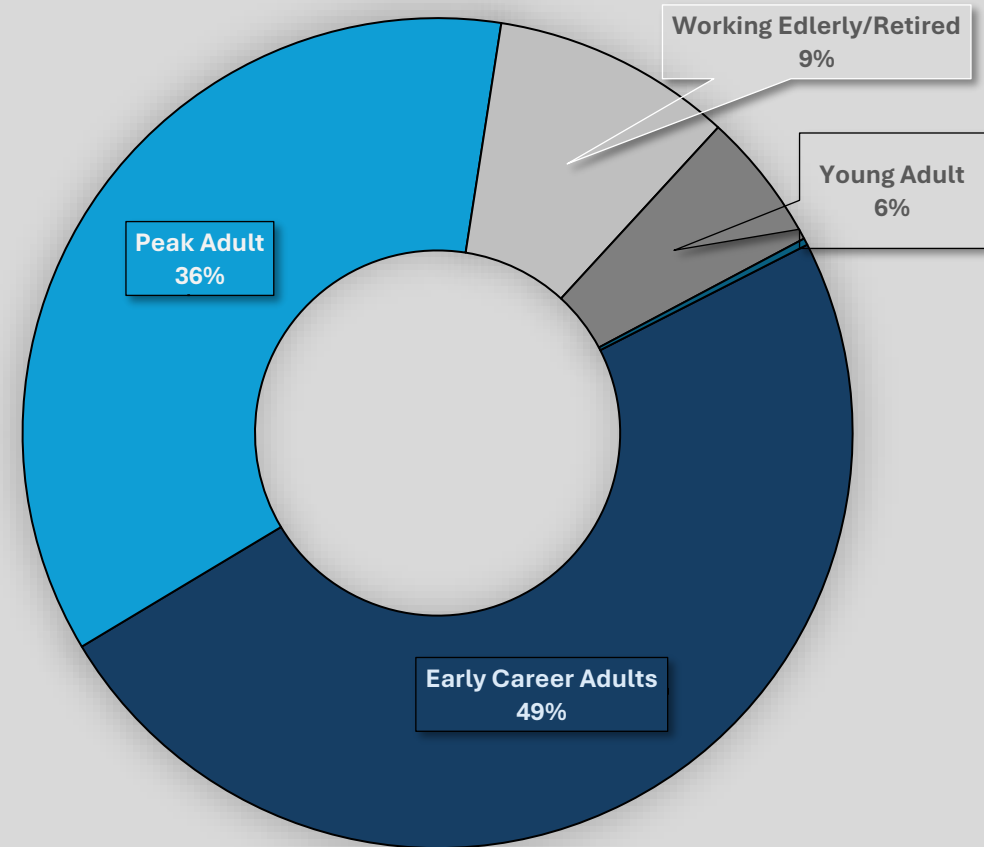
The post-winter Surge

- There is a huge surge of Casual users activating immediately after [Winter Season](#) ends, creating a targeted window for promotion.



Trips Demography

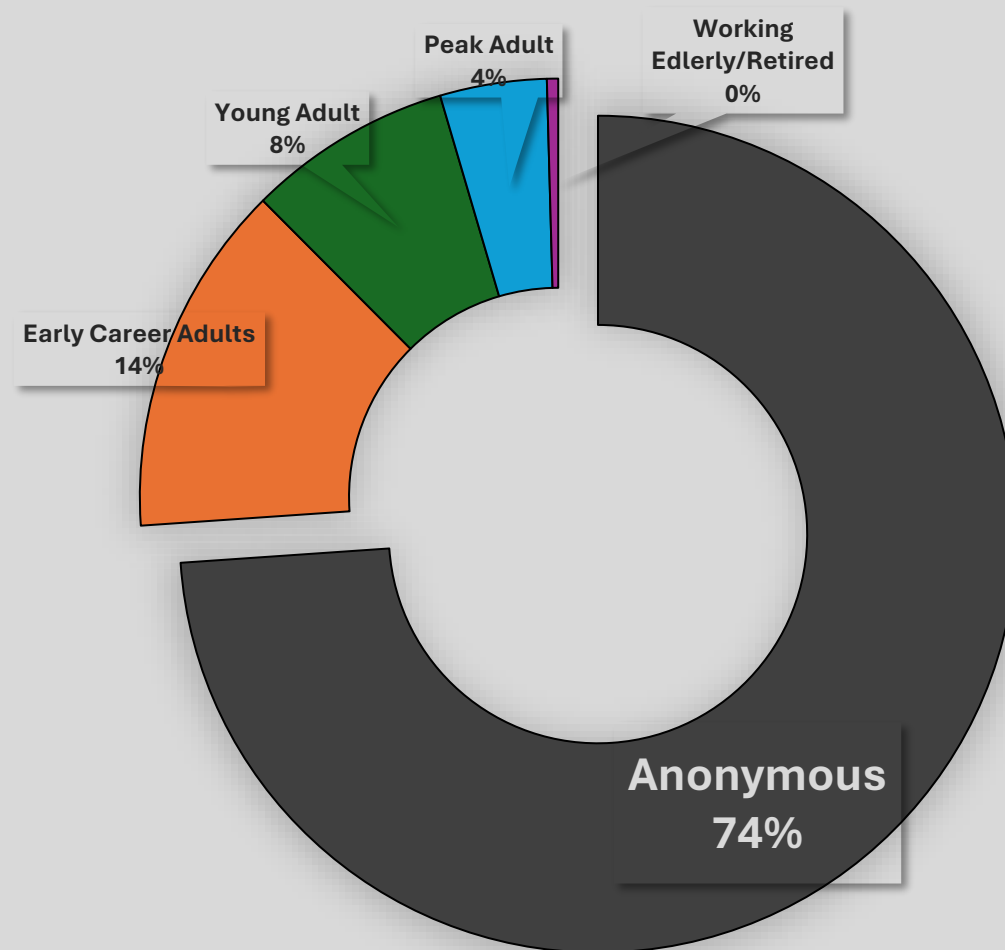
- As expected, working individuals utilize bikes for work day commute. Where can target annual members who use of bikes for point-to-point work commute, if we had any promotion for continues subscription.



Source: divvy_trips2019_Q1 age group for Members



However, we face a huge gap in birth year registration for casual riders. Without age and demographic markers, precision targeting is hindered.



Source: divvy_trips2019_Q1 age group for casual

Note : 2020 drop collecting birthyear on dataset

CONCLUSION

Bridging the Data Gap to Unlock Growth

The Core Limitation:

- Our behavioral analysis clearly shows *how* casual riders use Cyclistic differently; **longer, leisure-focused, weekend trips**. Preventing us from mapping them to annual member uses our bikes.
- There is **critical demographic data gap** exists, particularly among casual riders.

The Opportunity:

- Casual riders already love our ecosystem, keeping bikes out **3x longer than members** (34.6 mins vs. 11 mins) and surging in the spring. By incentivizing profile registration completion, we can convert anonymous casual riders into highly identifiable marketing targets.

RECOMMENDATION

Mapping Behavioral Reality to Strategic Action

Casual want leisurely , 30 plus minutes weekend ride



Create Leisure –Centric membership tier.

We could market an annual membership that emphasizes **unlimited long-duration weekend riding** or partners with scenic destinations to appeal to the "explorer" nature of casual riders

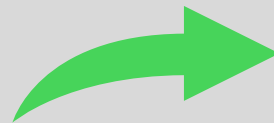
Casual ridership spikes massively in early spring



“Spring - Start” Seasonal Perks.

Capitalized the incoming season, capture the users at the exact moment their intent to ride is highest. Converting excitement into annual lock-in.

Casual user demographic data is missing



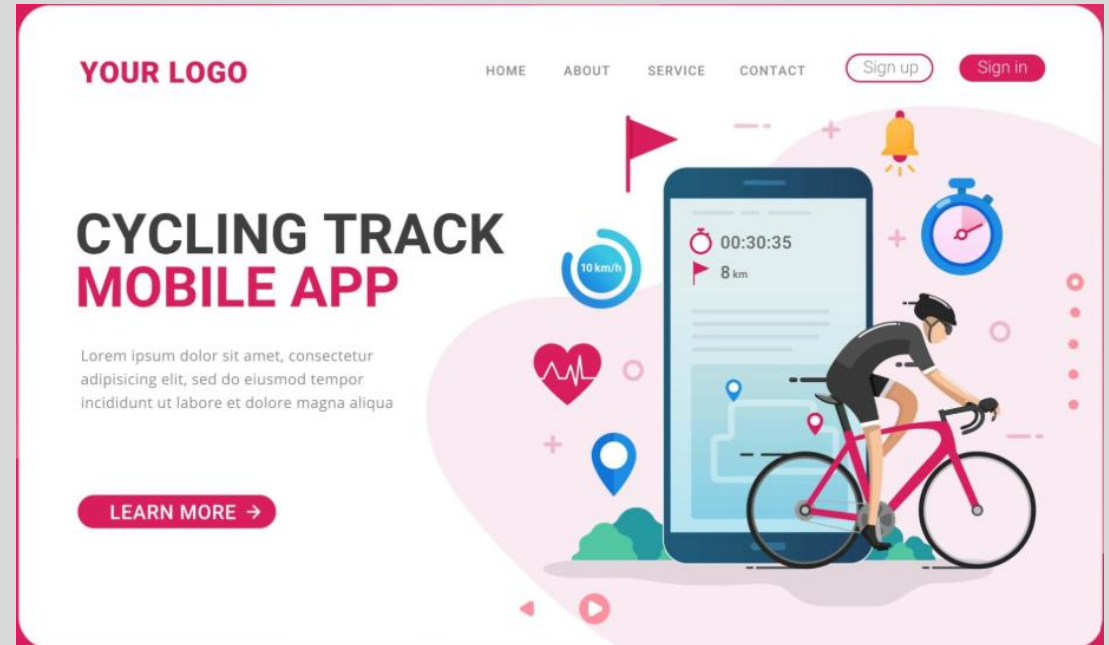
Incentivize Profile Completion.

offering a one-time ride discount in exchange for demographic information

For furthermore exploration (V2)

The Conversion Engine: “Premium Member-Exclusive Features.”

- A 35-minute ride is no longer just a commute, it is a legitimate **cardio workout**. Many casual riders are likely using Cyclistic for physical exercise, weekend cardio, or active exploration without even realizing their health impact.
- Instead of giving everyone the same basic map interface, gate the advanced health tracking tools behind the Annual Membership.



Cyclistic Premium Health Dashboard Concept. Source: apfDesign / Getty Images

- Thank you